

Customer Churn Prediction Using Artificial Neural Networks



A Deep Learning Approach to Predict and Analyze Customer Churn in Telecom Industry



Data Analysis



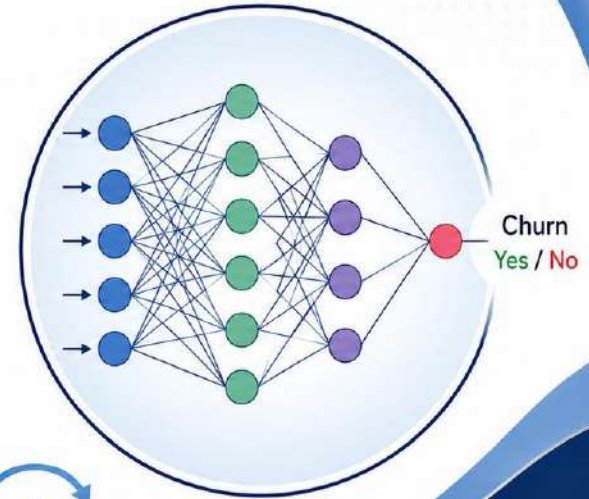
Artificial Neural Networks



Model Evaluation & Comparison



Insights & Prediction



DATASET

Telco Customer Churn Dataset

A dataset containing information about telecom customers and their subscription behavior, used to predict whether a customer will churn or not.



Objective:

Predict whether a customer is likely to **churn** (leave) or not.



Dataset Name

Telco Customer Churn Dataset



Domain

Telecommunications



Machine Learning Task

Binary Classification



Target Variable

Churn (Yes / No)



Total Records

7,043



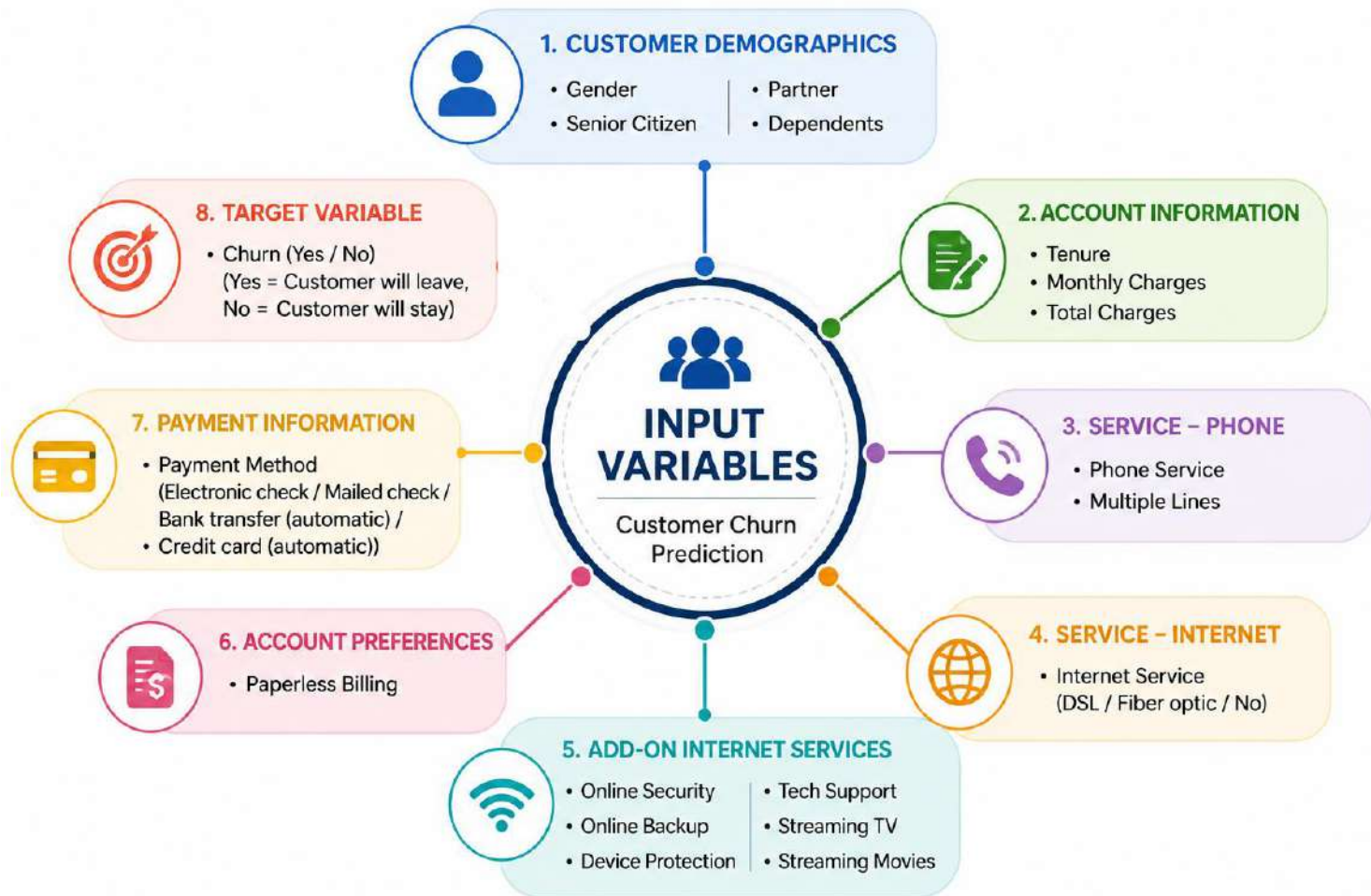
Original Features

21



Final Features after Encoding

26



SAMPLE DATASET

Customer Churn Prediction Dataset – Sample Records

CustomerID	Gender	Senior Citizen	Partner	Dependents	Tenure (Months)	Phone Service	Multiple Lines	Internet Service	Online Security	Online Backup	Device Protection	Tech Support	Streaming TV	Streaming Movies	Contract	Paperless Billing	Monthly Charges (USD)	Total Charges (USD)	Churn
7590-VHVEG	Female	0	Yes	No	1	Yes	No	DSL	No	Yes	No	No	No	No	Month-to-month	Yes	29.85	29.85	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	56.95	1889.5	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-month	Yes	53.85	108.15	Yes
7795-CFOCW	Female	0	No	Yes	45	Yes	No	DSL	Yes	No	Yes	Yes	No	No	One year	Yes	42.30	1840.75	No
9237-HQITU	Male	1	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	70.70	151.65	Yes
9305-CDSKC	Female	1	No	No	8	Yes	Yes	Fiber optic	No	No	No	Yes	Yes	Yes	Month-to-month	Yes	99.65	820.50	Yes
1452-KIOVK	Male	0	Yes	No	22	Yes	Yes	Fiber optic	Yes	Yes	No	No	Yes	Yes	One year	Yes	89.10	1949.40	No
6713-OKOMC	Female	0	No	Yes	10	Yes	No	DSL	Yes	No	No	Yes	Yes	Yes	Month-to-month	No	29.75	301.90	No
7892-POOKP	Male	0	Yes	Yes	28	Yes	Yes	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes	104.80	3046.05	Yes
6388-TABGU	Female	0	No	Yes	62	Yes	No	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	Yes	56.15	3487.95	No



This dataset contains 21 features describing customer demographics, services, and account information. The target variable "Churn" indicates whether the customer has churned (Yes) or not (No).



KEY FEATURES INCLUDE:

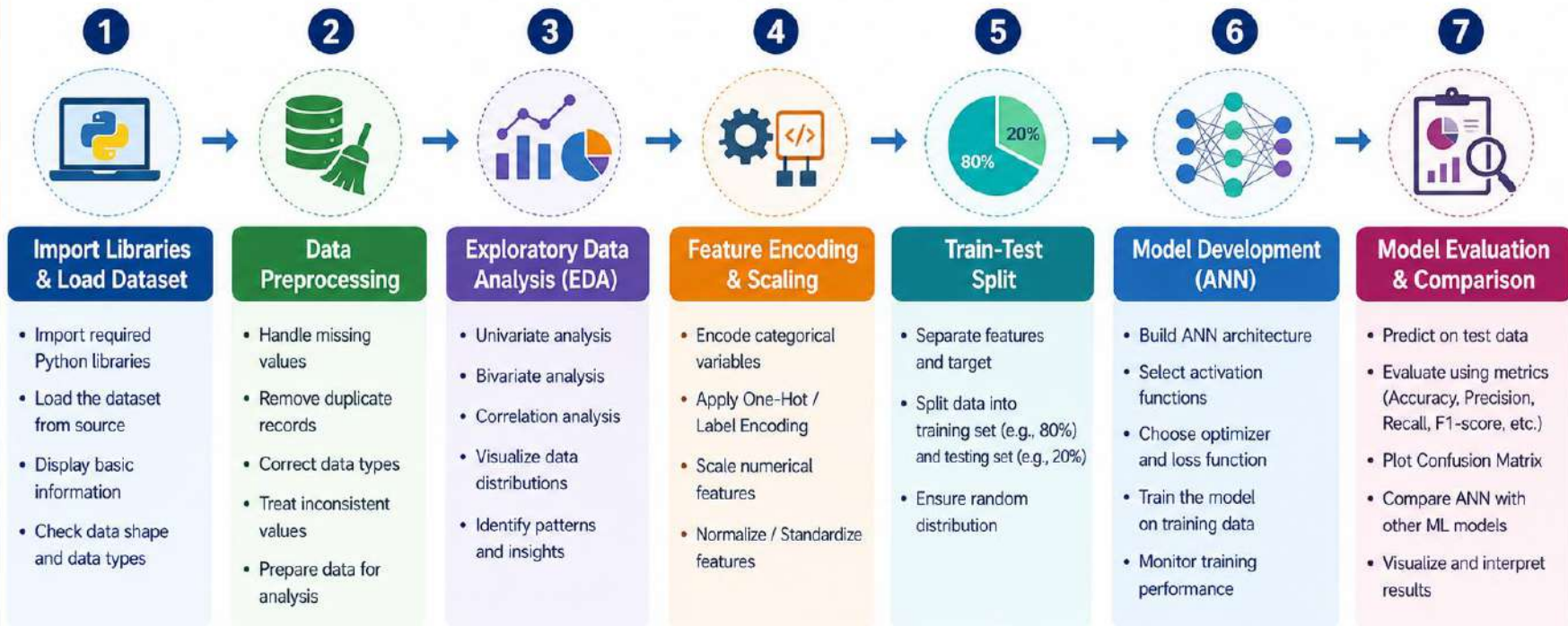
- Demographics (Gender, Senior Citizen, Partner, Dependents)
- Services (Phone, Internet, Online Security, Backup, etc.)
- Account Info (Tenure, Contract, Paperless Billing)
- Charges (Monthly Charges, Total Charges)
- Target (Churn)



PROCESS FLOW USING PYTHON



Machine Learning Project Workflow (Generalized)



KEY OUTCOME



Clean and prepared data



Useful insights from data



Well-trained ANN model



Performance evaluation



Model comparison and selection

Key ANN Hyperparameters – Tuning Options and Expected Effects

Parameter	Current Value	Possible Changes	Expected Effect
 Number of Hidden Layers	2	3–5 layers	Learn more complex patterns
 Neurons per Layer	26, 15	32, 64, 128, etc.	Increase model capacity
 Activation Function	ReLU	Leaky ReLU, ELU, Tanh	Better gradient flow
 Learning Rate	Default Adam	0.1, 0.01, 0.001, 0.0001	Faster or more stable convergence
 Optimizer	Adam	RMSprop, SGD+Momentum, Nadam, AdamW	May improve convergence
 Epochs	100	150–300	Better learning if underfitting
 Batch Size	Default (32)	16, 32, 64, 128	Influences training stability
 Dropout Rate	None	0.2–0.5	Reduces overfitting
 Regularization	None	L1, L2	Improves generalization
 Weight Initialization	Default	He, Xavier	Faster convergence
 Early Stopping	Not used	Monitor validation loss	Prevents overfitting
 Validation Split	Not used	20% validation	Better model selection